CIENCIAS DE LA COMPUTACIÓN
UNIVERSIDAD DE CHILE

COMPUTATIONAL APPROACHES TO LEXICAL SEMANTIC CHANGE DETECTION WITH WIC-BASED AND LLM-BASED SENSE INDUCTION

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Introduction

- language is constantly evolving, reflecting processes within both language and society

cell → small monastery cell, prison room

- later, it gained a new sense (telecommunications) in 1977

(Online Etymology Dictionary)

LSCD - Applications

Linguistics: reveals how social and cultural dynamics, cognitive processes,and lexical structure interact to shape word usage over time

Lexicography: tracks the emergence of new word senses and helps refine historical dictionaries

LSCD - Applications

Social and Political Science: investigates cultural and ideological shifts, the evolution of political discourse, and semantic signals tied to societal phenomena such as armed conflict

Digital Humanities: enables computational analysis of historical corpora across disciplines

Thesis Roadmap

1- Introduction

2 - Research Questions

3 - LSCD Spanish dataset

4 - WSI optimization for LSCD

5 - Enhancing LLMs for LSCD

6 - Conclusions

Lexical Semantic Change Detection

- **Binary Task**: Given a set of target words, determine which words have gained or lost senses between C1 and C2, and which have remained semantically stable.

- **Ranking task**: Rank a set of target words according to the degree of lexical semantic change between C1 and C2 by comparing their sense frequency distributions across the two corpora.

(Schlechtweg et al., 2020)

Lexical Semantic Change Detection

- **COMPARE:** Given two usages of a target word, this task focuses on estimating semantic change by considering pairs of usages in which one occurrence comes from the earlier corpus and the other from the later one.

DURel Relatedness Scale

4: Identical

3: Closely Related

2: Distantly Related

1: Unrelated

(Schlechtweg et al. 2018)

Example

- Usage A (1850): The prisoner was confined to a narrow cell in the tower.

- Usage B (2001): She forget her cell at home and missed the call.

- Judgment: 1 - Unrelated

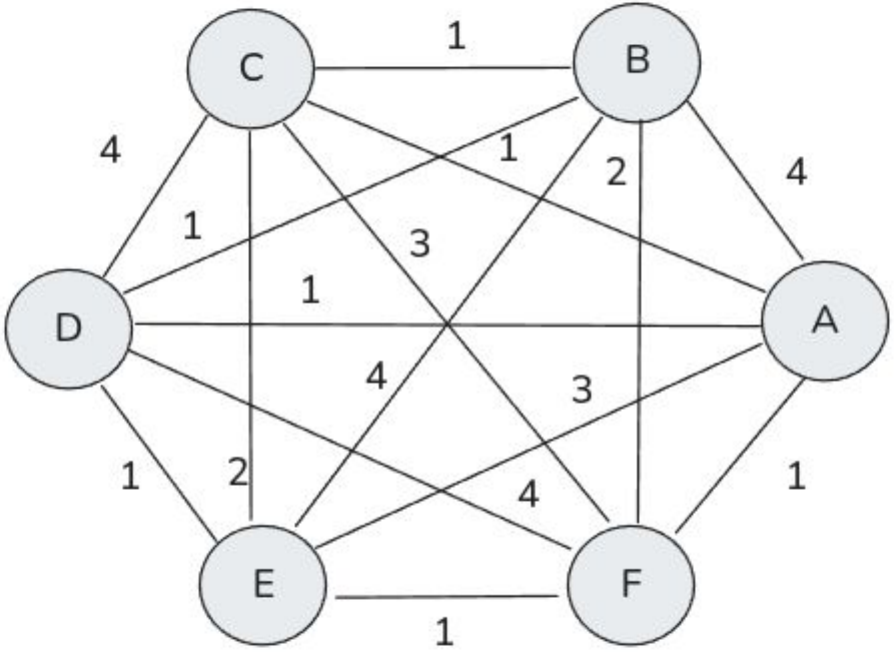
Word Usage Graph (WUG)

- $G = (U, E, W)$, weighted , undirected graph
- $u \in U$, represents word usages
- $w \in W$, represents semantic proximity of pair of usages $(u1,u2) \in E$

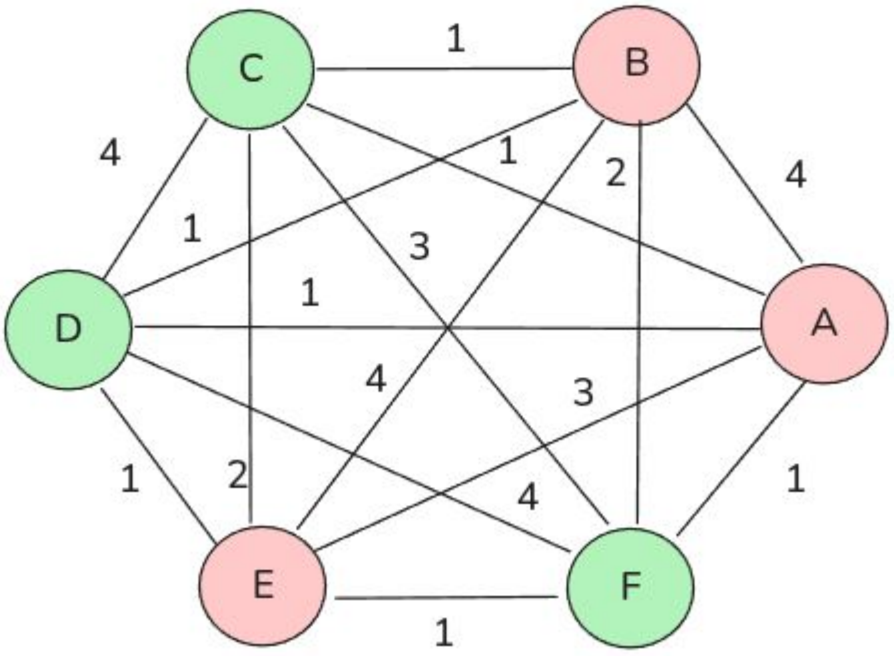
Example

ID	Period	Usage	Meaning
A	1810	Fui al banco a retirar dinero.	financial institution
B	1855	El banco me concedió un préstamo.	financial institution
E	1906	El banco central emitió un comunicado oficial.	financial institution
C	1994	Nos sentamos en el banco del parque a descansar.	bench
D	2008	El banco estaba mojado después de la lluvia.	bench
F	2020	Se quedó dormido en un banco de madera junto al río.	wooden bench

WUG



WUG



Background - SemEval 2020 Task 1

- unsupervised task
    - binary task
    - ranking task

Sense	C1 (historical)	C2 (modern)
chamber	12	4
biology	18	11
phone	0	18

(Schlechtweg et al. 2020)

Background - SemEval 2020 Task 1

- 4 datasets presented and a set of target words W={w1,w2,…,wn}

Language	C1	C2	n (target words)
English	1810 - 1860	1960 - 2010	37
German	1800 - 1899	1946 - 1990	48
Latin	-200 - 0	0 - 2000	40
Swedish	1790 - 1830	1895 - 1903	31

(Schlechtweg et al. 2020)

Background - SemEval 2020 Task 1

Main findings

- Type-based models outperformed token-based models - despite contextualized representations being theoretically more adequate to represent the different senses of a word

System type	Subtask 1	Subtask 2
Type-based systems	**.625**	**.329**
Token-based systems	.598	.258

Benchmarks beyond SemEval 2020 Task 1

- Italian dataset
    - Binary task
    - type-based systems dominated

- Russian dataset
    - 3 periods of time
    - COMPARE task
    - token-based systems outperformed type-based systems

An Open Challenge

- Inconsistent findings across languages

- Italian and Russian datasets used different annotation approaches

- No benchmark existed annotated with the same WUG methodology to test whether SemEval findings generalize beyond English and German

Extending LSCD to Spanish

- +450 million native Spanish speakers across +20 countries
- Spanish is underrepresented in NLP benchmarks generally
- Spanish is a highly polysemous language

Research questions

- Can we create a reliable Spanish dataset for the LSCD field?

- Can LSCD systems used in other datasets in different languages maintain their performance in the Spanish language?

DWUG ES - LSCD dataset

- Two diachronic corpora

Corpus	**Periods**	**Tokens**
Old	1810 - 1906	~13M
Modern	1994 - 2020	~22M

DWUG ES - LSCD dataset

- Largest LSCD dataset ~62K judgments from 12 native Speakers

Corpus	Periods	Tokens
Old	1810 - 1906	~13M
Modern	1994 - 2020	~22M

DWUG ES - LSCD dataset

- 100 annotated target words (20 development, 60 evaluation, 20 discarded)

Corpus	Periods	Tokens
Old	1810 - 1906	~13M
Modern	1994 - 2020	~22M

Annotation process and Agreement

- 12 native Spanish speakers: 4 Chilean, 4 Colombians, 2 Cubans, 1 Spaniard, 1 Venezuelan

- 20 usages samples per word per corpus (40 usages in total)

Annotation process and Agreement

- WUG annotation using DURel 4-point scale

- Correlation clustering - sense frequency distributions - gold labels

Annotation process and Agreement

Partition	Words	Krippendorff’s α
Development	20	.53
Evaluation	60	.58
Discarded	20	.27
Full	**100**	**.53**

Annotator disagreement by nationality

Modern usage (C2)

“La investigación ha demostrado que las plantas de maíz …”

-> meaning: biological organism (crop plants)

Historical usage (C1)

“nada se tocaría a la planta y ornamentación de la fábrica árabe …”

-> meaning: architectural layout

Colombian	Spaniard	Cuban	Chilean
1	4	4	3

LSCDiscovery: A shared task in LSCD Spanish

- It was organized a shared task: Mandatory phases

Phase	Task	Description	Metric
Phase 1	Graded Change Discovery	Rank all vocabulary words (4385)	Spearman's ρ
Phase 2	Binary Change Detection	Classify pre-selected words as change/not changed	F1

LSCDiscovery: A shared task in LSCD Spanish

- It was organized a shared task: Optional tasks

Task	**Description**
Graded Change Detection	Same as Discovery but on pre-selected annotated words
COMPARE	Predict the negated DURel score
Sense Gain Detection	Identify only words that **gained** senses
Sense Loss Detection	Identify only words that **lost** senses

LSCDiscovery: A shared task in LSCD Spanish

Main characteristics

- Fully unsupervised
- 6 teams participated in Phase 1, and 7 teams in Phase 2

Novelty

- LSCDiscovery introduced the Discovery scenario
  - a more realistic evaluation requiring systems to cover the full vocabulary, as lexicographers would need to do in practice

LSCDiscovery: A shared task in LSCD Spanish

Results - Phase 1

Rank	System	Representation	Approach	Spearman's ρ
1	GlossReader	Token (XLM-R)	APD	.735
2	DeepMistake	Token (XLM-R)	APD	.702
3	HSE	Token (BERT)	Clustering	.553
—	Baseline 1	Type (SGNS)	Cosine Distance	.543

LSCDiscovery: A shared task in LSCD Spanish

Key findings and conclusions

1. Token-based systems dominate

2. Performance for graded change comparable to previous shared tasks

LSCDiscovery: A shared task in LSCD Spanish

Key findings and conclusions

3. APD has a clear upper bound

    a. COMPARE task is an approximation of the Graded Change Task

	Correlation with JSD
Gold COMPARE	.92

4. Clustering-methods among the best-performing systems for the first time

    b. important because current systems exploit correlation between change measures and do not model annotation procedure

LSCDiscovery - Insights

- Aggregation-based approaches, such as APD are approaching a performance ceiling

- LSCDiscovery showed that clustering-based systems appeared among top-performers - suggesting that approaches which more closely model the sense-level structure underlying the annotation procedure may provide advantages for semantic change detection

LSCDiscovery - Insights

- However, the relationship between WSI performance and LSCD performance has never been systematically studied

Research question

Does optimization for WSI transfer to LSCD and optimization in LSCD transfer to WSI?

Example

Period   Usage

A (1810) Fui al banco a retirar dinero. (financial institution)

B (1855) El banco me concedió un préstamo. (financial institution)

E (1906) El banco central emitió un comunicado oficial. (financial institution)

C (1994) Nos sentamos en el banco del parque a descansar. (bench)

D (2008) El banco estaba mojado después de la lluvia. (bench)

F (2020) Se quedó dormido en un banco de madera junto al río. (wooden bench)

Hypothesis

Optimizing clustering configurations for WSI will improve their downstream performance on LSCD.

Annotation Procedure

Problem

- Human annotation is expensive - annotators judge pairwise usages, but each judgment contributes to only one target word
- LSCDiscovery required ~62K judgments for just 100 words
- Scaling to full vocabulary is unfeasible with human annotators

Annotation Procedure

Problem

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Solution

Automate the annotation procedure using Word-in-Context (WiC) models.

WiC Task

Given two usages of the same target word, the task is to determine whether they have the same meaning.

WiC Models

- Automate what human annotators do in the DURel framework - replacing human judgments with model predictions

- Annotation procedure scalable to any numbers of words

WiC Model - DeepMistake

- Fine-tuned XLM-R model trained to judge semantic proximity between usage pairs

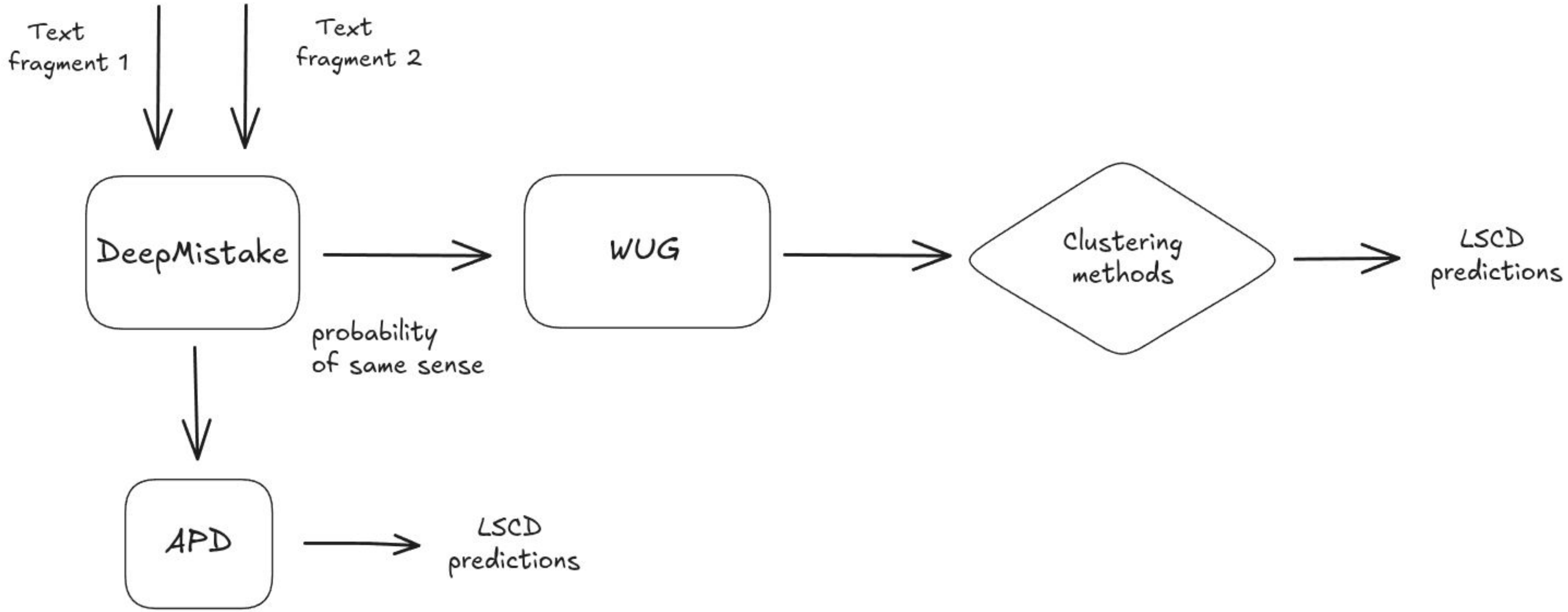
- Its output (probability of same sense) replaces human DURel judgments as edge weights in the WUG

(Arefyev et al., 2021)

Datasets

Dataset	Id	n	C1	C2
German	DWUG DE	24	1800 - 1899	1946 - 1990
Spanish	DWUG ES	60	1810 - 1906	1994 - 2020
English	DWUG EN	37	1810 - 1860	1960 - 2010

Experimental setup



WUG

- complete graph generated -> [0.0 - 1.0]
    - each language

- hyperparameters
    - threshold t (edges below t are cut them out)
    - binarize (weighted edges -> [0, 1])
    - normalized (-> [0, 1])
    - use disconnected edges (-> [True, False])
    - quantile (-> [1,..., 10])
    - fill_diagonal (-> [True, False])

Clustering methods

Clustering method	Parameter	Values
Chinese Whispers	weighting	[lin, top, log]
Correlation Clustering	threshold_cc	0.5
WSBM	distribution	[binomial, poisson, discrete, normal, exponential]
Spectral Clustering	number of clusters	[Silhouette, Calinski-Harabas, Eigengap]

German results

Clustering method	Optimized - WSI		Optimized - LSCD	
	ARI	Spearman	ARI	Spearman
Chinese Whispers	.622 ± .130	.768 ± .241	.602 ± .078	.768 ± .241
Correlation clustering	.622 ± .132	.873 ± .110	.666 ± .096	.715 ± .207
WSBM	.710 ± .068	.911 ± .156	.613 ± .158	.828 ± .169
Spectral Clustering - Silhouette	.771 ± .090	.873 ± .141	.715 ± .067	.835 ± .130
Spectral Clustering - Calinkski - Harabasz	.755 ± .087	.834 ± .077	.702 ± .087	.774 ± .111
Spectral Clustering - Eigengap	.633 ± .144	.807 ± .202	.426 ± .208	.596 ± .213

Spanish results

Clustering method	Optimized - WSI		Optimized - LSCD	
	ARI	Spearman	ARI	Spearman
Chinese Whispers	.475 ± .049	.590 ± .260	.453 ± .066	.498 ± .254
Correlation clustering	.412 ± .113	.661 ± .106	.411 ± .115	.615 ± .121
WSBM	.444 ± .022	.553 ± .186	.480 ± .078	.602 ± .243
Spectral Clustering - Silhouette	.351 ± .108	.578 ± .173	.371 ± .112	.420 ± .187
Spectral Clustering - Calinkski - Harabasz	.363 ± .097	.420 ± .239	.316 ± .113	.301 ± .271
Spectral Clustering - Eigengap	.310 ± .084	.595 ± .152	.301 ± .094	.567 ± .189

English results

Clustering method	Optimized - WSI		Optimized - LSCD	
	ARI	Spearman	ARI	Spearman
Chinese Whispers	.243 ± .098	.700 ± .209	.256 ± .129	.598 ± .198
Correlation clustering	.192 ± .104	.307 ± .194	.080 ± .076	.293 ± .189
WSBM	.226 ± .144	.562 ± .297	.045 ± .025	.555 ± .214
Spectral Clustering - Silhouette	.279 ± .095	.600 ± .205	.088 ± .034	.505 ± .233
Spectral Clustering - Calinkski - Harabasz	.106 ± .027	.157 ± .425	.059 ± .042	.288 ± .215
Spectral Clustering - Eigengap	.064 ± .038	.395 ± .207	.054 ± .050	.126 ± .232

APD vs Custering-based approach

Dataset	Best WSI -> LSCD	APD
German	.873 ± .141	.760 ± .121
Spanish	.590 ± .260	.653 ± .250
English	.600 ± .205	.638 ± .202

Finding

- WSI optimization transfers to LSCD across all three datasets, despite differences in absolute performance.
- LSCD-optimized configurations do not consistently transfer back to WSI

However, sense-based clustering does not consistently outperform APD:

- German: Clustering > APD
- Spanish and English: APD > clustering results

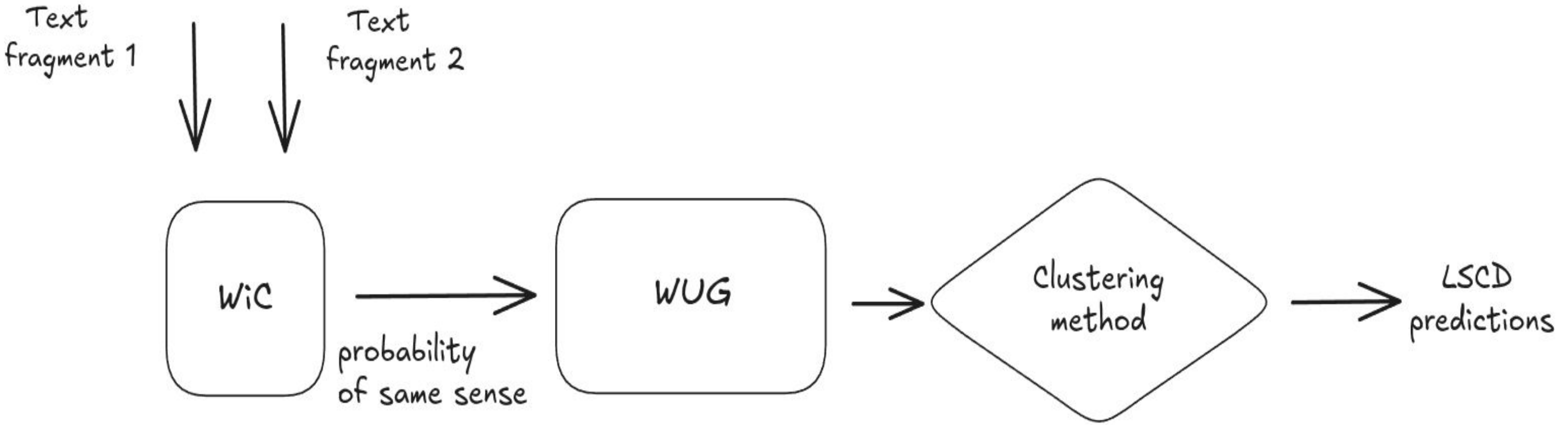
Finding

Hyperparameter analysis

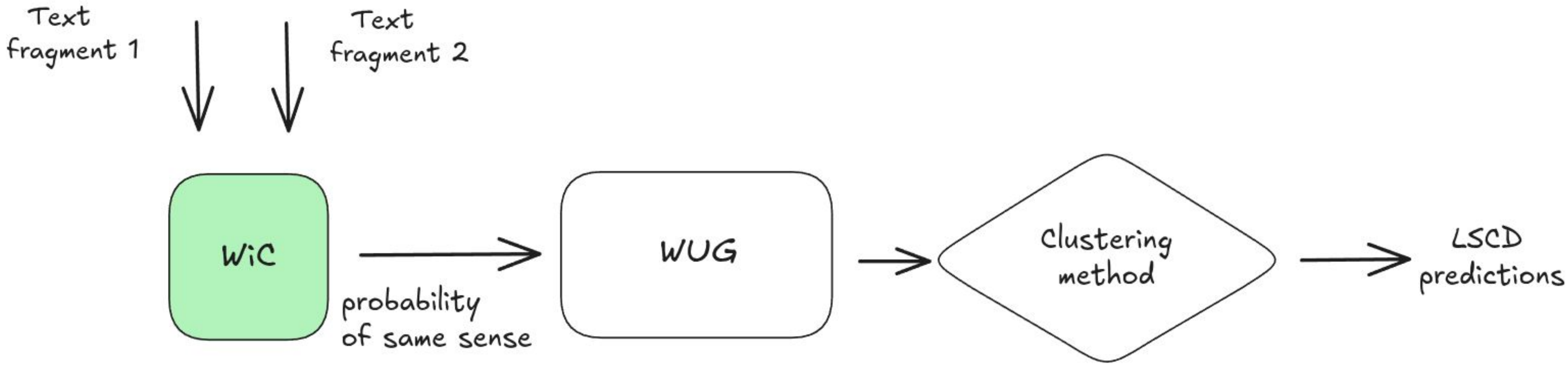
- No single configuration was optimal across datasets and clustering methods.

- Hyperparameter effects were dataset- and method-dependent.

LSCD pipeline



LSCD pipeline



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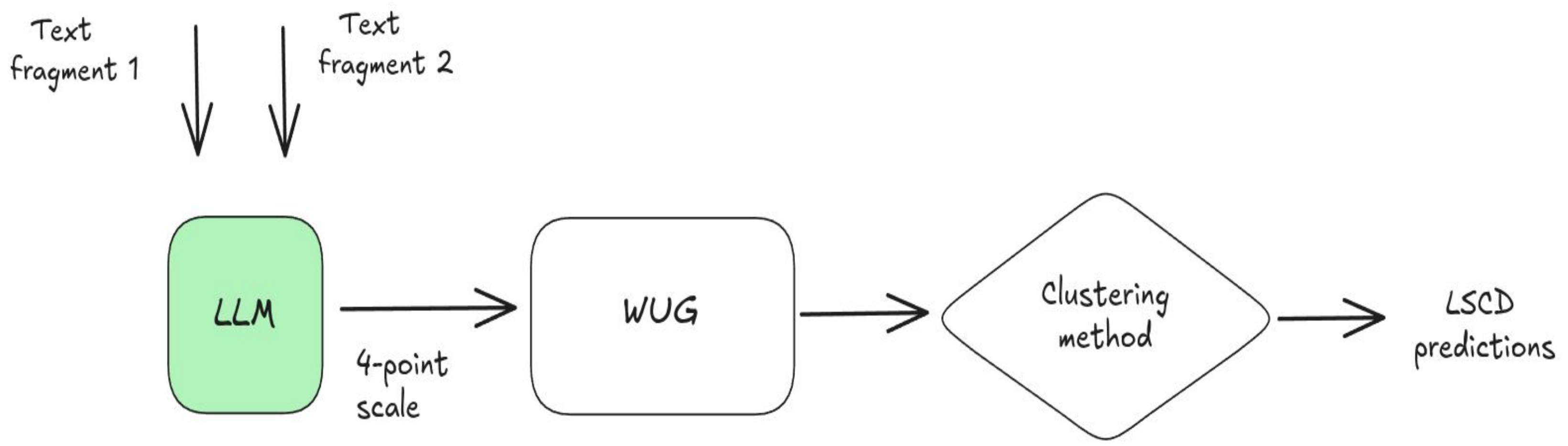

Research question

Can Large Language Models compete with specialized models in LSCD?

- Can automatically optimized prompts yield better results for the LSCD task than manually crafted prompts designed through prompt engineering?

- Can LLMs solve the Graded Change LSCD task well? Can these results surpass the WiC models reported as state-of-the-art?

- Can LLMs outperform state-of-the-art LSCD models at the annotation level?

Hypotheses

- Automatically optimized prompts improve the quality of LLM-generated semantic proximity judgments and, consequently, LSCD performance.

- LLMs can generate semantic proximity judgments that are competitive with those produced by specialized WiC models.

Metrics

- SPR_WiC - Spearman correlation between the annotations provided by human annotators and models.

- SPR_LSCD - Spearman correlation between gold graded change scores and model predictions

Models

- LLMs
  - Llama 3.1 (8B)
  - Mixtral 8x7B (Jiang et al., 2024)
  - Llama 3.3 (70B)
- WiC models
  - DeepMistake (Arefyev et al., 2021)
    - SOTA for Russian and Spanish datasets
    - MCL, enMCL, MCL-> es (fine-tuned on various WiC datasets across multiple languages)
  - XL-LEXEME (Casotti et al., 2023)
    - SOTA for English, Swedish, German datasets

Clustering methods

- Spectral clustering (von Luxburg, 2007)

- Agglomerative clustering (Jain and Dubes,1988)

- Weighted Stochastic Block Model (Peixoto, 2019)

Clustering methods

<table>
<tr><th>Clustering method</th><th>Parameter</th><th>Values</th></tr>
<tr><td>WSBM</td><td>distribution</td><td>[binomial, poisson, discrete, normal, exponential]</td></tr>
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<tr><td>Agglomerative Clustering</td></tr>
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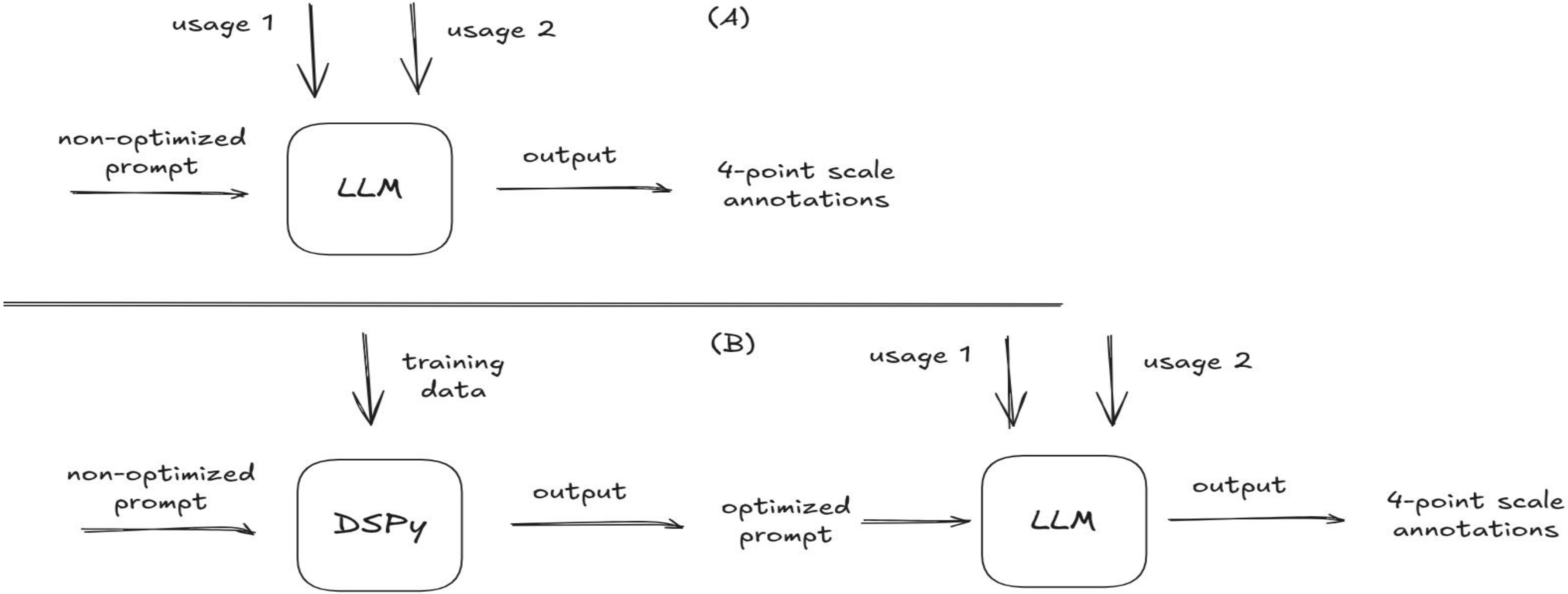
Experimental Setup - LLMs



Prompts

- optimized prompt from Yadav et al. (2024)
  - translate the prompt into Spanish
  - extend the prompt using samples from the DURel framework (Schlechtweg et al., 2018)

- optimized prompts further using DSPy framework (Khattab et al., 2023)

Prompt optimization

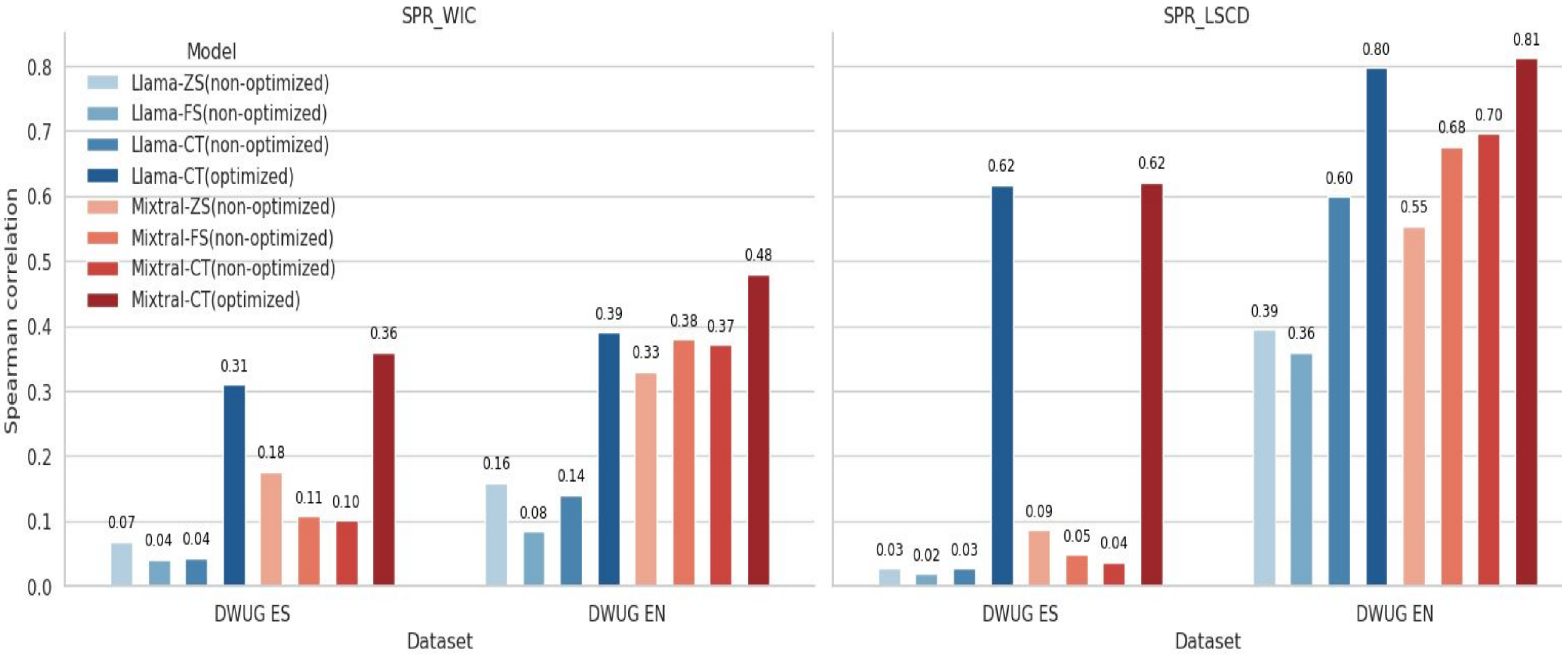
Accuracy

Dataset - prompt	Llama 3.1		Mixtral		Llama 3.3	
	NOP	OP	NOP	OP	NOP	OP
DWUG ES - PrS	26.8	29.5	32.6	31.22	32.33	39.77
DWUG ES - PrE	26.5	35.5	33.7	34.80	40.88	46.33
DWUG EN - PrS	25.75	31.0	32.8	40.75	37.25	45.5
DWUG ES - PrE	28.75	37.25	33.25	38.75	35.75	49.25

NOP - non optimized prompt, OP - optimized prompt

PrE - prompt written in English, PrS - prompt written in Spanish

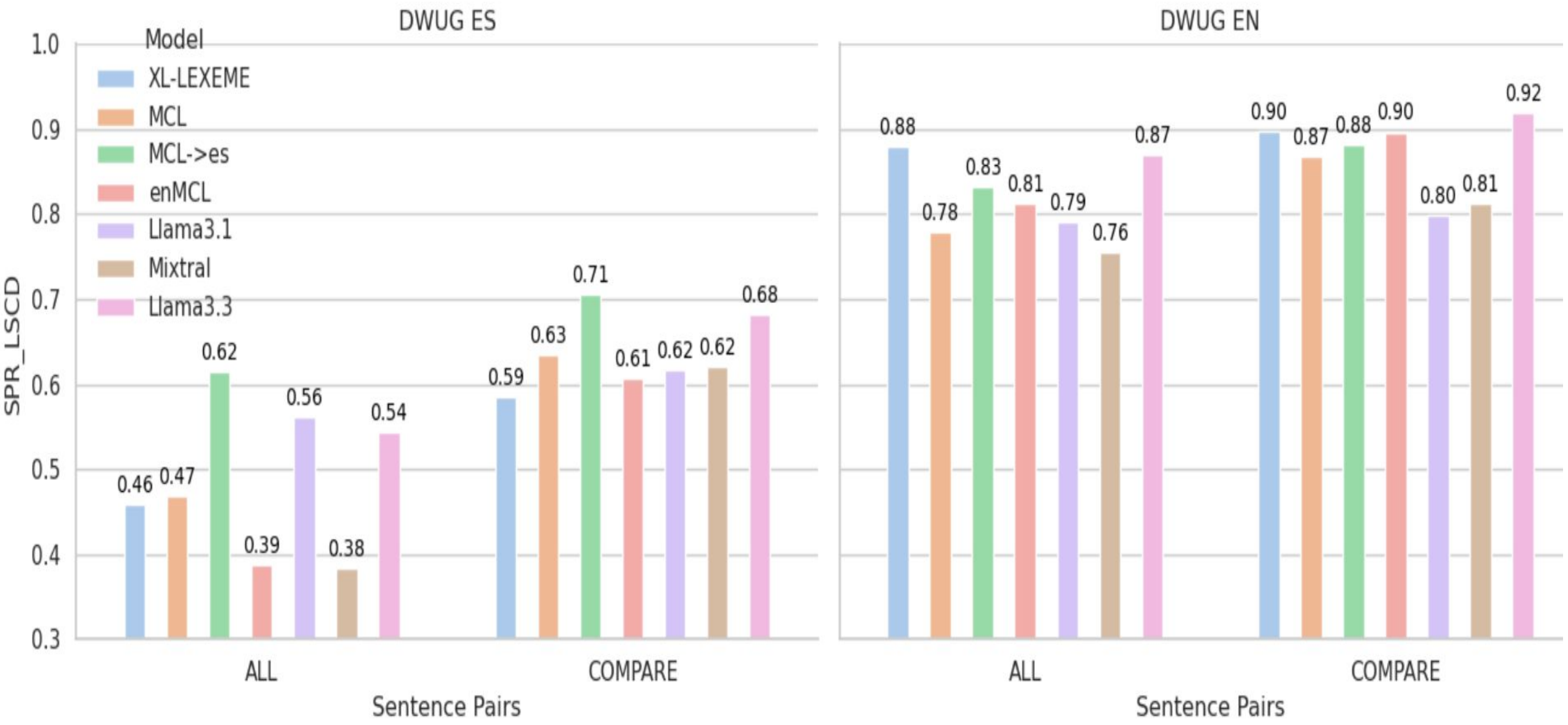
Non-optimized Prompts vs. Optimized Prompts



RQ2

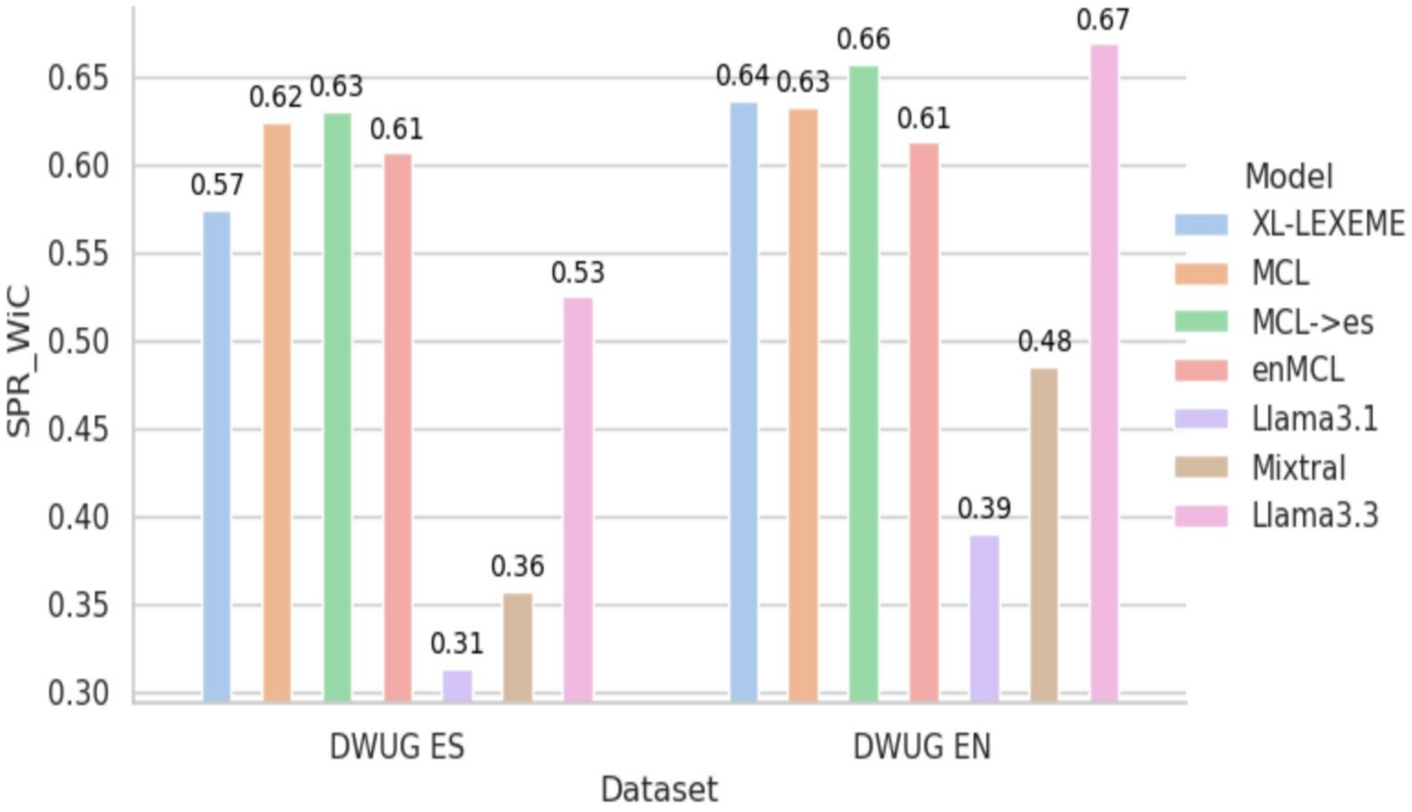
- Can LLMs solve the Graded Change LSCD task well? Can these results surpass the WiC models reported as state-of-the-art?

Specialized WiC models vs LLMs in SPR_LSCD



- Can LLMs outperform state-of-the-art LSCD models at the annotation level?

Specialized WiC models vs LLMs in SPR_WiC



Findings

- recent prompt optimization techniques are crucial for achieving better results on the Graded Change LSCD task, as demonstrated by the performance of Llama3.3:70B

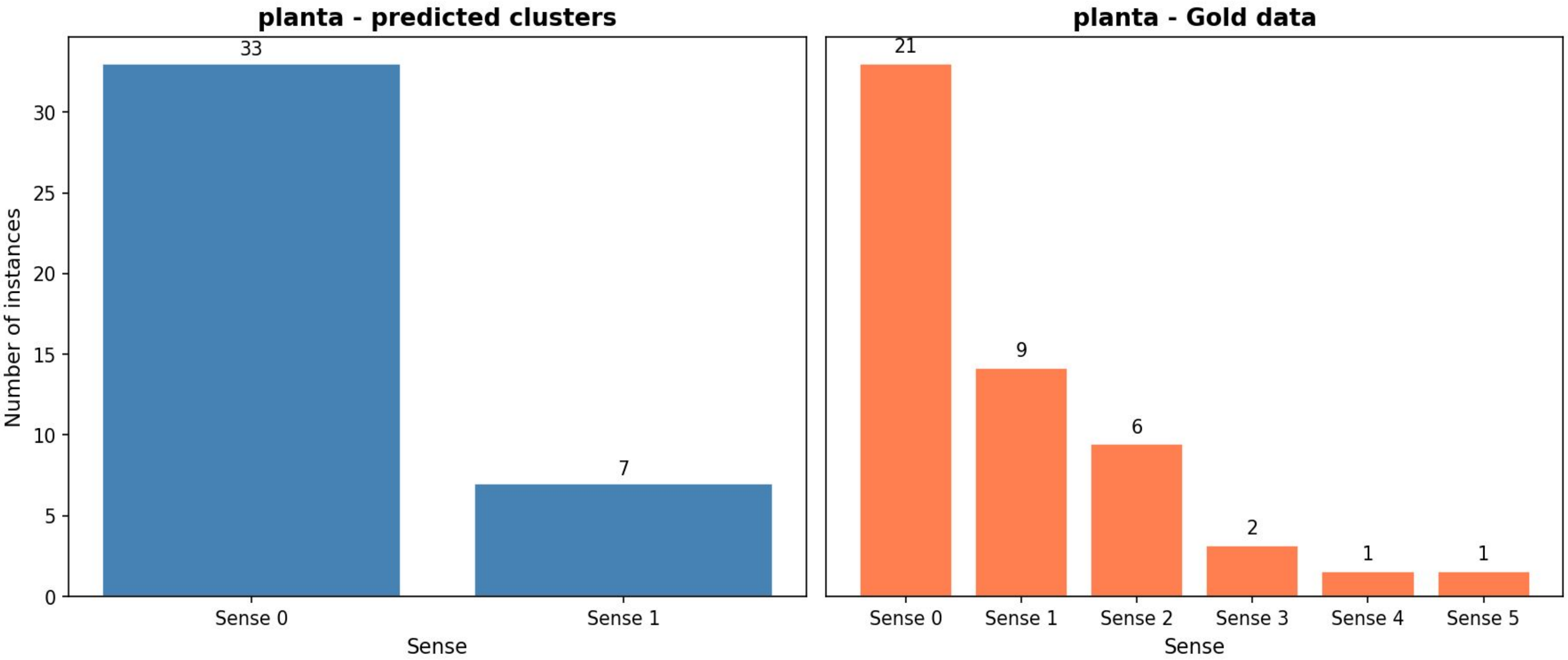
- medium-sized LLMs such as Mixtral:8x7B and Llama3.1:8B still underperform compared to smaller and faster specialized LSCD models in both the DWUG EN and DWUG ES datasets
  - in addition to optimization techniques, the size of the model also significantly influences the results

- LLMs performance on LCSD is strongly language-dependent - competitive results in English do not generalize to Spanish

Qualitative analysis

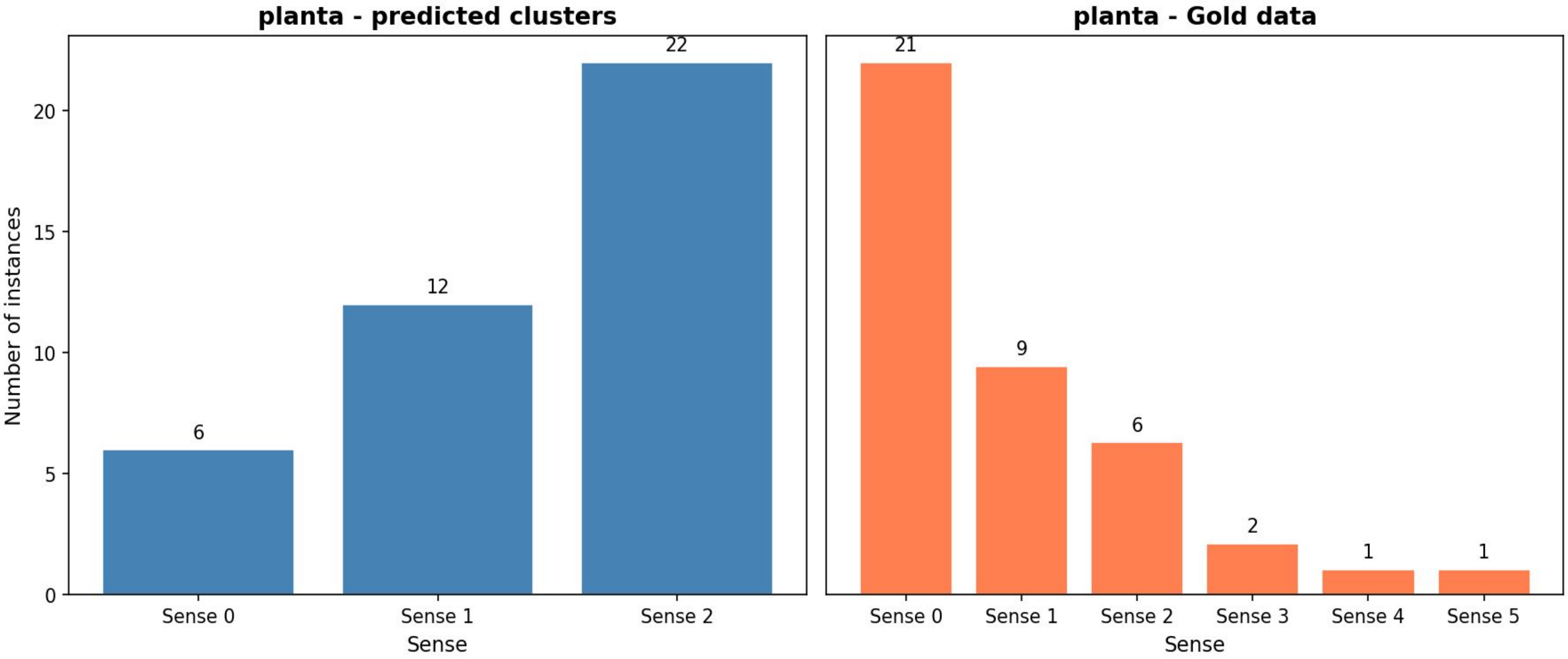
- Quantitative evaluation identifies which model performs best
- It does not explain how the semantic change is represented
- We therefore inspect the induced senses

Qualitative analysis



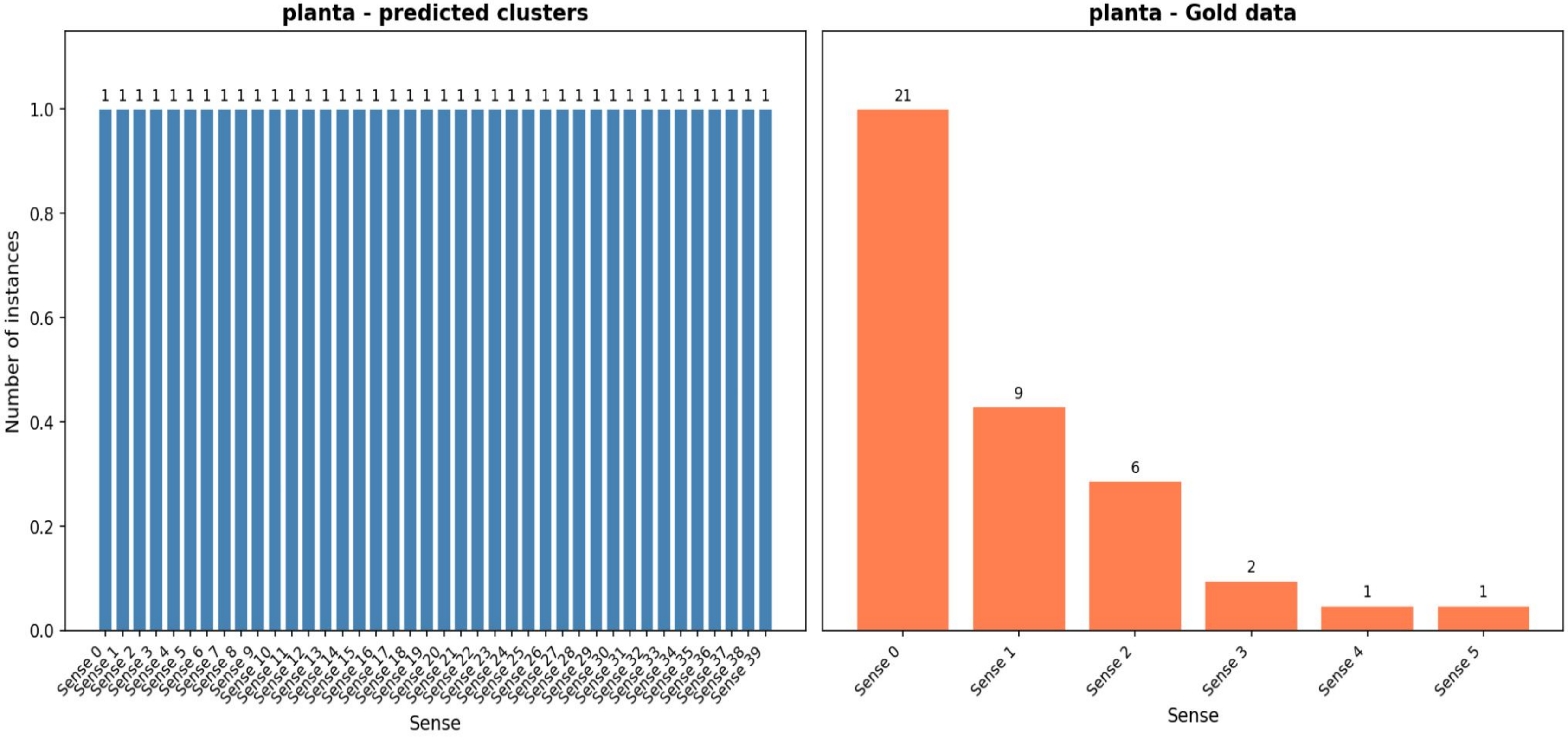
Spectral Clustering + Llama3.3:70B annotations

Qualitative analysis



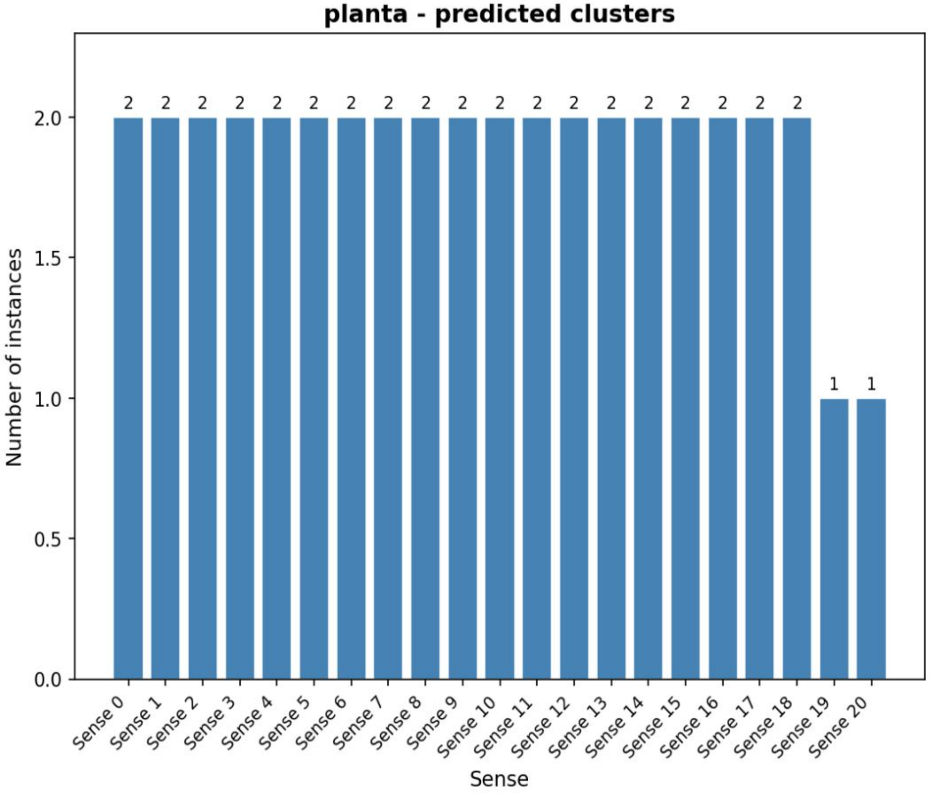
Spectral Clustering + MCL->es annotations

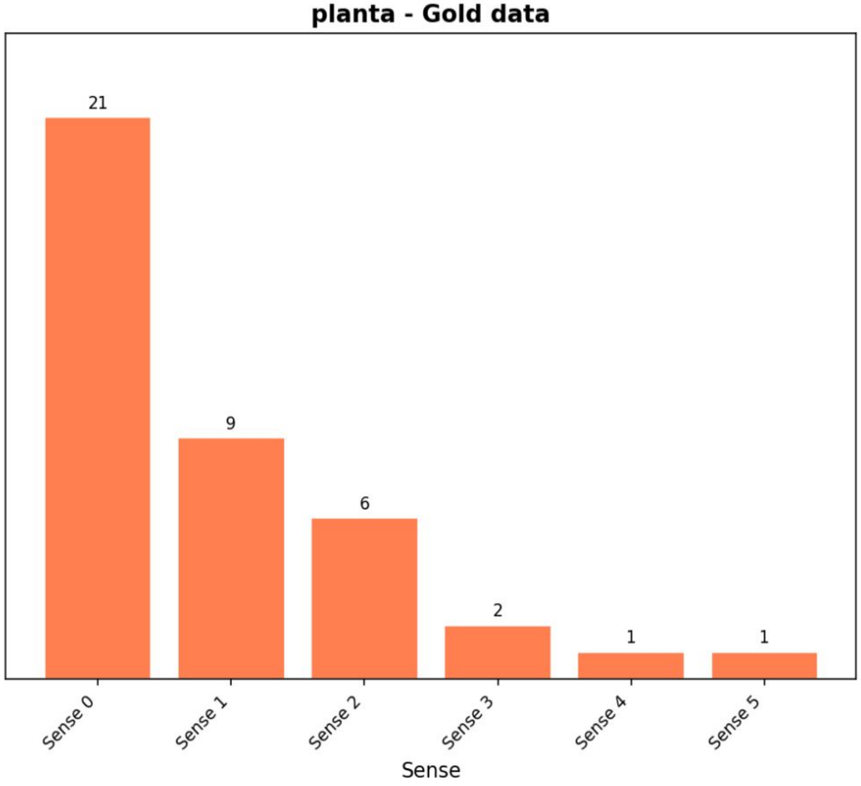
Qualitative analysis



WSBM + Llama3.3:70B annotations

Qualitative analysis





WSBM + MCL->es annotations

Findings - Qualitative analysis

- Different clustering methods induce substantially different sense inventories.
- Models with comparable LSCD performance may recover different semantic structures.
- Graded semantic change scores alone do not reveal the quality of the induced sense inventories.

Contributions

- First LSCD dataset and shared task for Spanish - extending research beyond predominantly studied languages
- Clustering-based and WiC-based approaches are most effective - highlighting the central role of sense-related information in LSCD

Contributions

- First systematically study of the relationship between WSI and LSCD performance

- WSI-optimized configurations consistently transfer to LSCD - but LSCD-optimized configurations do not transfer back to WSI

Contributions

- Automatic prompt selection consistently outperforms manual prompt design across models and languages for the LSCD task

- LLMs approximate human judgments in English - but remain less reliable than specialized models in Spanish

Conclusion

LSCD can be understood as a sense-driven task whose successful modeling depends on representations that capture semantic structure across time

Conclusion

- Three studies converge on the same insight: sense-level structure is the key for LSCD

- LLMs offer promising capabilities under automatic prompt optimization - but do not yet provide a universal solution

Conclusion

- Progress in LSCD will rely on the integration of sense-aware modeling, principled evaluation, and interpretability-driven analysis

- Strong quantitative scores do not guarantee linguistically coherent sense clusters - interpretability matters

Limitations

- LLM experiments limited to English and Spanish - German excluded due to computational and budgetary cost

- Pretrained models primarily trained on contemporary text - may under represent historical language usage

Limitations

- The experiments involving LLMs rely on a finite set of model architectures and prompting strategies. The evaluated models were constrained by computational and financial costs, which limited the inclusion of substantially larger frontier-scale models

Limitations

- The annotation schemes adopted in this work at the graded annotation level, necessarily impose structural assumptions that may influence model behaviour as well as the evaluation outcomes for LSCD

Limitations

- Due to the inherently non-deterministic nature of LLM inference, exact replication of results may be affected by stochastic decoding and model updates over time

Future work

- Meaning interpretability - develop methods to validate whether induced clusters and LLM representations correspond to linguistically meaningful senses, combining clustering analysis, probing techniques, and human-in-the-loop evaluation

(Giulianelli et al., 2023; Fedorova et al., 2024a; Fedorova et al.,2024b; Fedorova et al., 2025)

Future work

- Investigate whether substantially larger models exhibit stronger transfer capabilities for LSCD and semantic proximity modeling, and whether they close the performance gap in Spanish

Future work

- Explore dynamic topic models as a way to model sense distributions across time without requiring pairwise annotations, potentially addressing the granularity limitations observed in clustering

Motivation

- Sense-aware LSCD methods first induce word senses and then compare their distributions across time.
- The quality of the induced sense structure may therefore directly affect semantic change detection.
- However, the relationship between WSI performance and LSCD performance has not been systematically established.

Conclusion

- Three studies together show
  - dataset creation
  - WSI optimization to transfer parameters to LSCD task
  - LLM evaluation